There is only one Polish ring topology on the entire functions

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Fundamentals

Notation

- For any open subset $\Omega \subseteq \mathbb{C}$, $\mathcal{A}(\Omega)$ denotes the space of all holomorphic functions on Ω.
- $\mathcal{A}(\Omega)$ has the usual compact-open topology/topology of uniform convergence on compact subsets.
- $\mathbb{D} = \{\lambda \in \mathbb{C} : |\lambda| < 1\}$
- For $\alpha \in \Omega$, $M_\alpha = \{f \in \mathcal{A}(\Omega) : f(\alpha) = 0\}$.
- $\mathcal{I} = \{f \in \mathcal{A}(\Omega) : (\forall \alpha \in \Omega)[f(\alpha) \neq 0]\}$.
- For any ring R, $R^* = R \setminus \{0\}$.

Lemma (application of A. Atim, R.R. Kallman 2012)

Let G be a multiplicative Polish group with unity. If H and K are analytic subgroups of G so that $G = HK$ and $H \cap K = \{1\}$, then both H and K are closed.

Lemma

For a commutative Polish ring R with unity,

$$\mathcal{I}_R := \{x \in R : (\exists y \in R)(xy = 1_R)\}$$

is a Borel set.

Lemma (application of D. Montgomery 1936)

Let R be a commutative Polish ring with unity. Then the map $x \mapsto x^{-1}$, $\mathcal{I}_R \to \mathcal{I}_R$, is continuous if and only if $\mathcal{I}_R$ is a G_δ subset of R.

Particularly, if $\mathbb{F}$ is a Polish field, then $x \mapsto x^{-1}$, $\mathbb{F}^* \to \mathbb{F}^*$, is continuous.

Lemma (L. Bers 1948)

For any countable dense $\Lambda \subseteq \mathbb{C}$ and open connected $\Omega \subseteq \mathbb{C}$,

$$f \text{ is constant} \iff (\forall \lambda \in \Lambda)[f - \lambda \in \mathcal{I} \cup \{0\}].$$

Lemma

Let R be a Polish ring and $\phi : R \to \mathcal{A}(\Omega)$ be a ring isomorphism. Then

- $\phi^{-1}[\mathbb{C}]$ is closed,
- $\phi^{-1}[M_\alpha]$ is closed, and
- both $\phi^{-1}[\Omega]$ and $\phi^{-1}[\mathbb{C} \setminus \Omega]$ are Borel.

Theorem (C., Kallman)

Let R be a Polish ring and $\phi : R \to \mathcal{A}(\Omega)$ be an isomorphism of rings. If $\phi \restriction_{\phi^{-1}[\mathbb{C}]}$ is $\mathscr{B}\mathscr{P}$-measurable, then ϕ is continuous.

The Entire Functions

Theorem (Weierstrass 1894/5)

For a non-zero holomorphic function f, its set of zeros has no limit points. Conversely, given a countable set E with no limit points, there exists a holomorphic function so that its zero set is E.

Definition

Let R be a ring. We say that $x \in R$ is **algebraically wandering** if the map $\omega \to R$ defined by $n \mapsto x^{n+1}$ is injective. Let

$$\mathbf{W}(R) = \{x \in R : x \text{ is algebraically wandering}\}.$$

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Observe that the notion of algebraically wandering is an algebraic invariant.

Lemma

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Lemma

Let $W^+ = \{\lambda \in \mathbf{W}(\mathbb{C}) : |\lambda| > 1\}$ and notice that

$$\lambda \in W^+ \iff (\exists f \in \mathcal{A}^*(\mathbb{C}))(\forall n \in \omega)[f(\lambda^{n+1}) = 0].$$

Theorem (C., Kallman)

If R is a Polish ring and $\phi : R \to \mathcal{A}(\mathbb{C})$ is a ring isomorphism, then ϕ is a homeomorphism.

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If R is a Polish ring and $\phi : R \to \mathcal{A}(\mathbb{C})$ is a ring isomorphism, then ϕ is a homeomorphism.

- Let $\mathbb{K} = \phi^{-1}[\mathbb{C}]$.
- Distinguish the following projections:
  - $\pi : (R \times \mathbb{K})^2 \to R \times \mathbb{K}$ defined by $\pi(x, a, y, b) = (x, a)$
  - $p : (R \times \mathbb{K})^2 \to \mathbb{K}$ defined by $p(x, a, y, b) = b$
  - $P : R \times \mathbb{K} \to \mathbb{K}$ defined by $P(x, a) = a$

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  - $P : R \times \mathbb{K} \to \mathbb{K}$ defined by $P(x, a) = a$
- Define $A = \{\langle x, a, y, b\rangle \in (R \times \mathbb{K})^2 : x = (\phi^{-1}(z) - a)y + b\}$ and notice that $\pi \restriction_A$ is a bijection.

- Also define

$$B = \{\langle x, a\rangle \in R^* \times \mathbf{W}(\mathbb{K}) : (\forall n \in \omega)[p \circ (\pi \restriction_A^{-1})(x, a^{n+1}) = 0]\}$$

and observe that

$$P[B] = \phi^{-1}[W^+] = \phi^{-1}[\{\lambda \in \mathbb{C} : |\lambda| > 1\}]$$

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- Since $\mathbb{K}$ is a Polish field, $x \mapsto x^{-1}$, $\mathbb{K}^* \to \mathbb{K}^*$, is continuous so $\phi^{-1}[\mathbb{D}]$ is an analytic set.

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Meromorphic Functions

Notation

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Observe that $\mathcal{M}^*(\Omega)$ is a multiplicative group.

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Theorem (C., Kallman)

If $\mathcal{A}(\Omega)$ has a unique Polish ring topology, then $\mathcal{M}(\Omega)$ admits no Polish field topology.

Definition

For $\alpha \in \Omega$, define $\phi_\alpha : \mathcal{M}^*(\Omega) \to \mathbb{Z}$ by

- $\phi_\alpha(f) = 0$ if $f(\alpha) \neq 0$ and does not have a pole at α
- $\phi_\alpha(f) = n$ for $n > 0$ if f has a zero of order n at α
- $\phi_\alpha(f) = -n$ for $n > 0$ if f has a pole of order n at α

Observe that each ϕ_α is a homomorphism of groups.

Theorem (R.M. Dudley 1961)

Let H be a normed group with the discrete topology and G be a Polish group. Then any group homomorphism $\phi : G \to H$ is continuous.

Note that $\mathbb{Z}$ is a normed group.

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- for $\alpha \in \Omega$, $\mathcal{A}^*(\Omega) \cap \phi_\alpha^{-1}(1)$ is a neighborhood of $z - \alpha$.

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Bounded Functions on the Disk

Notation

Let

- $B(\mathbb{D})$ be the space of bounded holomorphic functions on $\mathbb{D}$ and
- H^∞ be $B(\mathbb{D})$ endowed with the topology of uniform convergence (compatible with the sup norm metric).

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- $B(\mathbb{D})$ be the space of bounded holomorphic functions on $\mathbb{D}$ and
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Theorem (C., Kallman)

$B(\mathbb{D})$ admits no Polish ring topology.

Lemma

H^∞ is complete metrizable but not separable.

This follows from the fact that

$$\left\{ \exp\left(\frac{z+\lambda}{z-\lambda} \right) : |\lambda| = 1 \right\}$$

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is a continuum sized relatively discrete subset of H^∞.

Lemma

As before, $M_\alpha = \{f \in B(\mathbb{D}) : f(\alpha) = 0\}$ is a principally generated maximal ideal.

Lemma (H.L. Royden 1958)

Let $\Lambda \subseteq \mathbb{C}$ be a countable dense set. Then

$$f \text{ is constant} \iff (\forall \lambda \in \Lambda)(\exists g \in B(\mathbb{D}))[g^2 = f - \lambda].$$

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$$f \text{ is constant} \iff (\forall \lambda \in \Lambda)(\exists g \in B(\mathbb{D}))[g^2 = f - \lambda].$$

Lemma

For any $\lambda \in \mathbb{C}$, $|\lambda| > 1 \iff z - \lambda \in \mathcal{I}_{B(\mathbb{D})}$.

Assume that $B(\mathbb{D})$ is a Polish ring topology.

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Future Work

We have shown that $\mathcal{A}(\Omega)$ has a unique Polish ring topology for all open $\Omega \subseteq \mathbb{C}$ that are conformally or anti-conformally equivalent to any of the following:

- $\mathbb{C}$
- Any connected open set so that its complement is bounded
- Any connected open subset so that its complement has non-empty interior
- Disconnected open sets

We have yet to show that the result holds for all open $\Omega \subseteq \mathbb{C}$.

Thank You!